

Supplementary Material: Cost-Effective Clinical AI for Dementia Progression Forecasting

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Supplementary Figures and Table

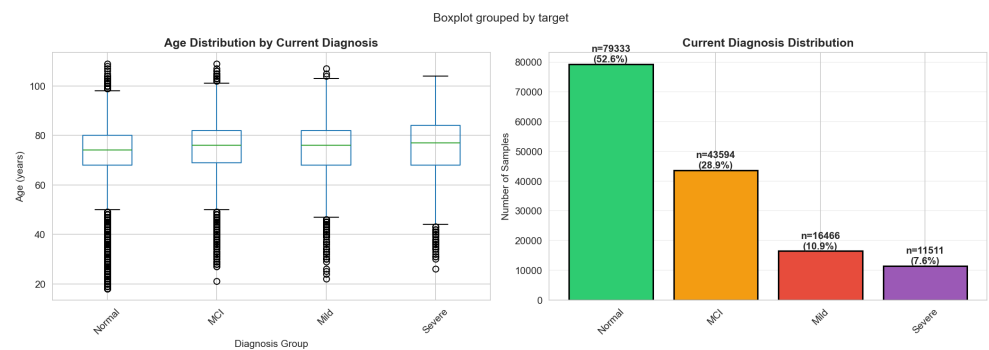


Figure S1: Demographic characteristics of the study cohort. Left: Age distribution stratified by current diagnosis. Right: Class distribution showing natural imbalance.

Table S1: Ablation Study: Model Complexity vs Performance.

Configuration	Accuracy	F1-Macro	Complexity
Base XGBoost	0.8859	0.8557	Low
XGBoost + SMOTE	0.8818	0.8486	Medium
Full Ensemble + SMOTE	0.8704	0.8390	High

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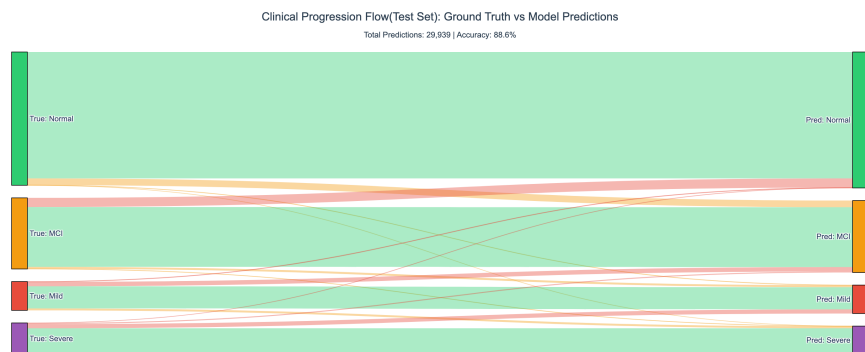


Figure S2: Sankey Diagram of Clinical Progression (Test Set). Flow widths are proportional to patient counts. Left: True diagnosis at current visit. Right: Model-predicted diagnosis at subsequent visit.

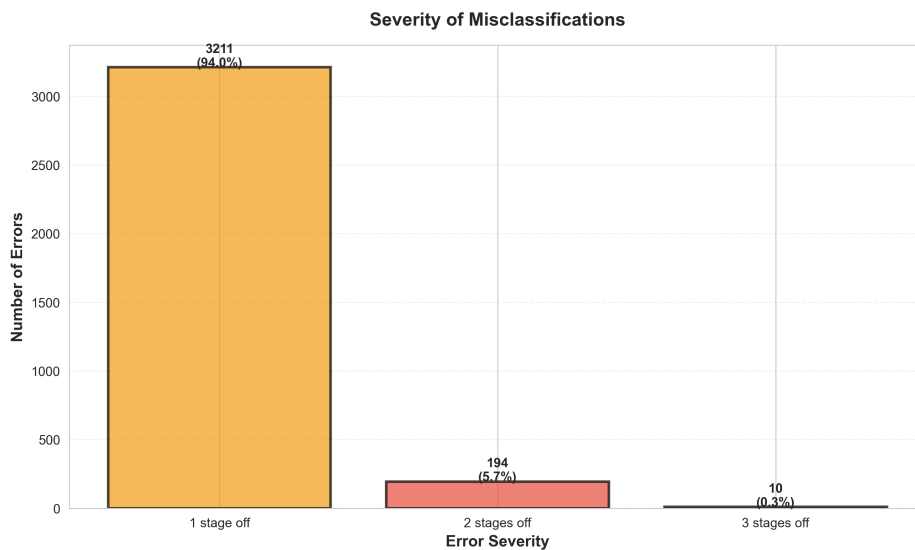


Figure S3: Error Severity Analysis. Most errors are one stage off, indicating clinically proximal predictions.

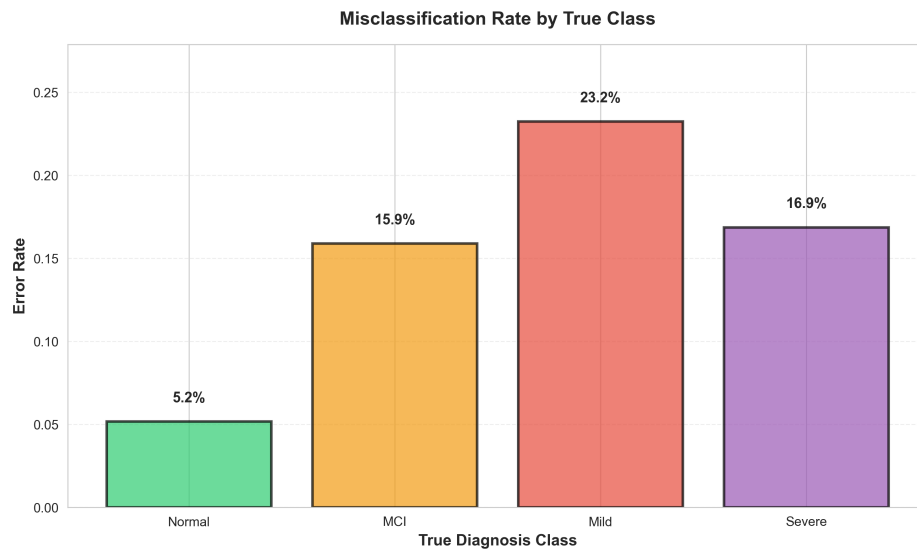


Figure S4: Error Rate by True Class. The Mild dementia class has the highest observed error rate.

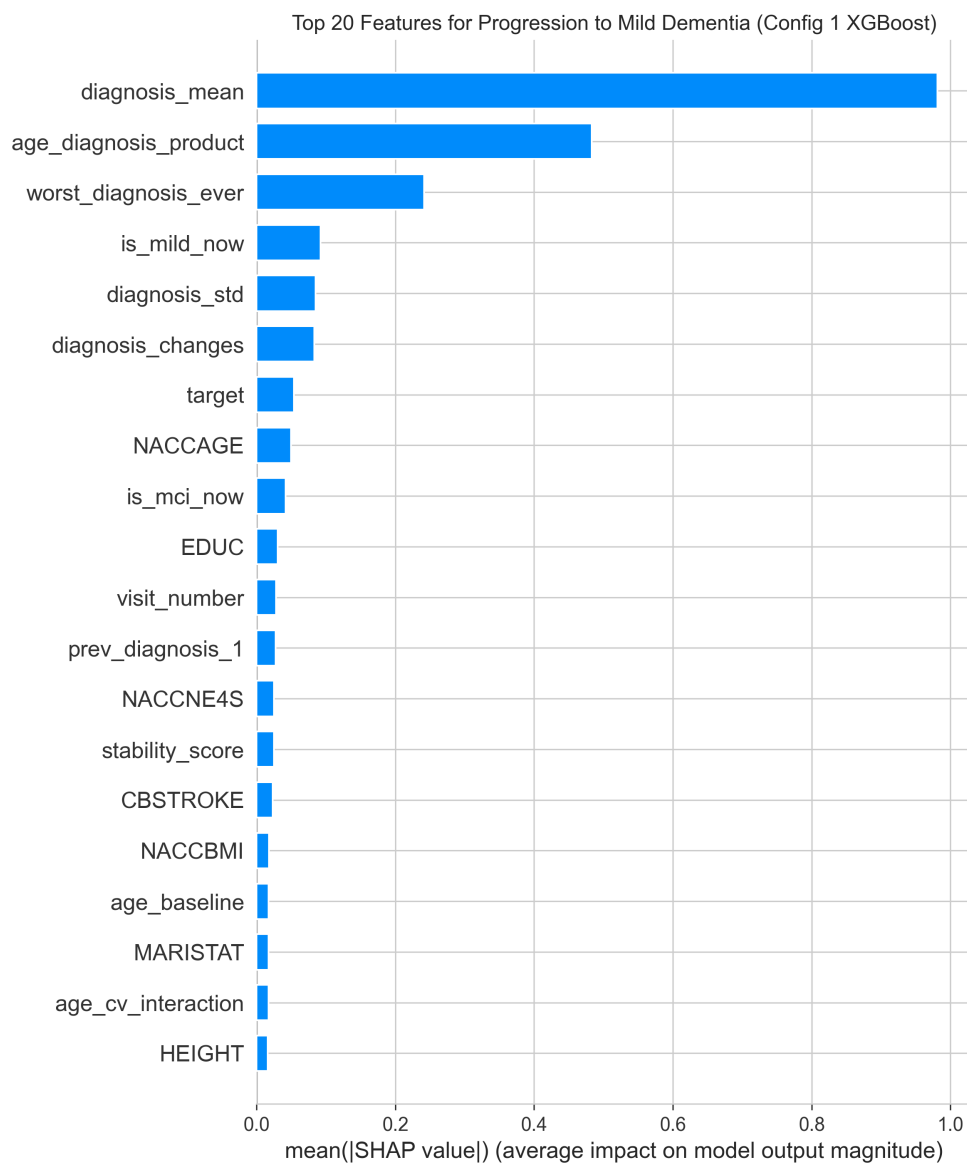


Figure S5: SHAP Feature Importance Bar Chart. Mean absolute SHAP values quantify each feature's contribution to model predictions.

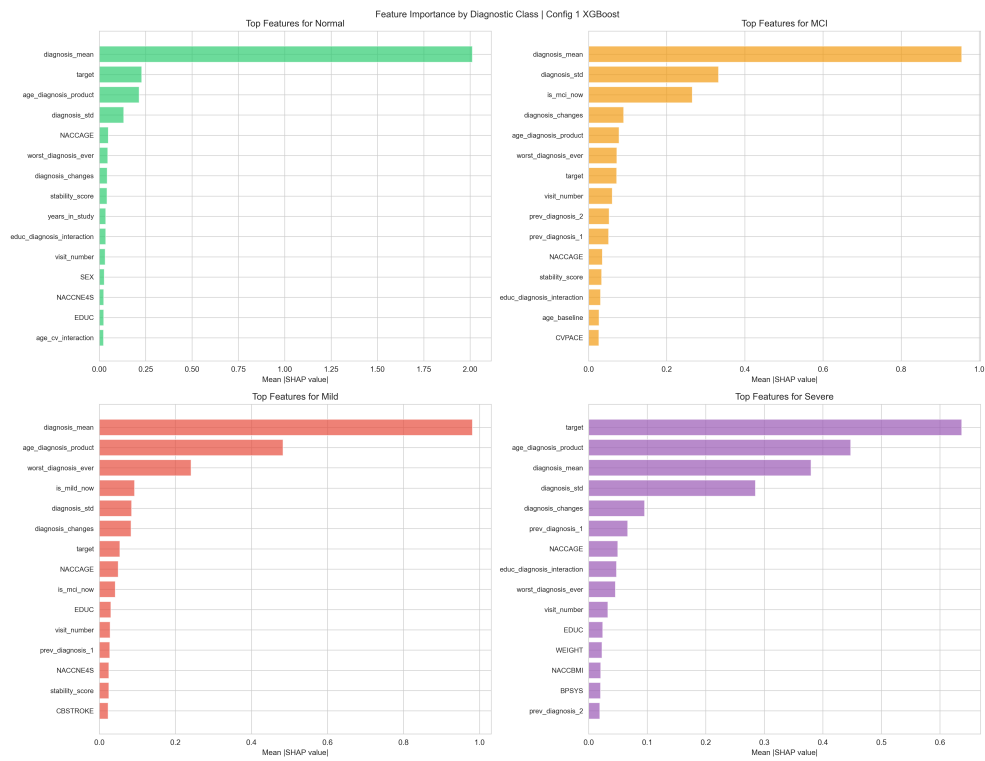


Figure S6: SHAP Feature Importance by Diagnostic Class. Each subplot shows the top contributing features for a diagnostic class.

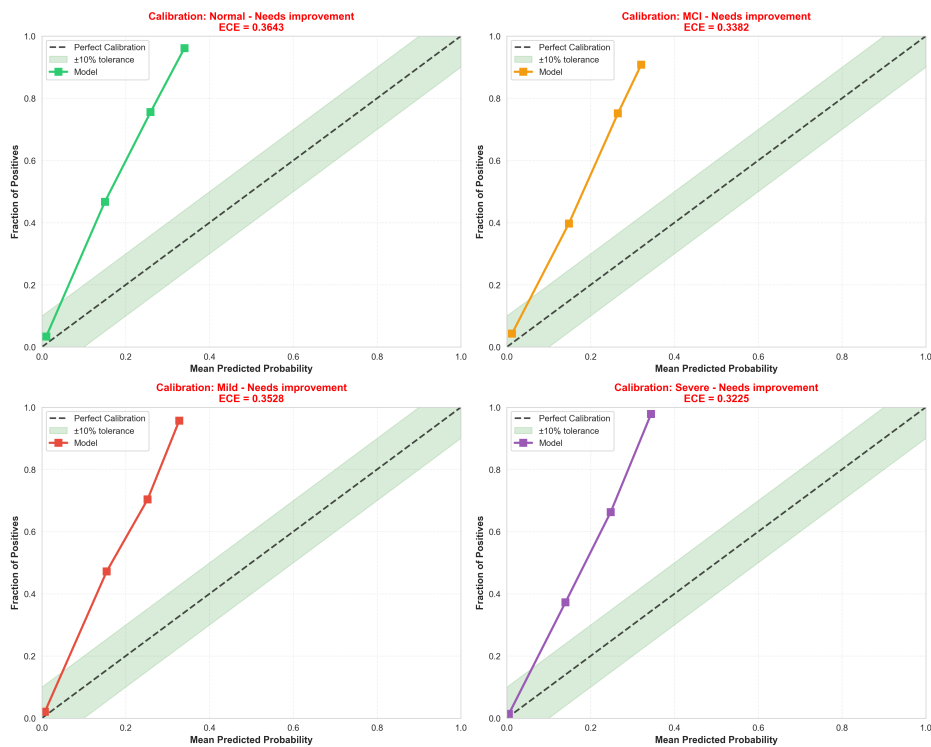


Figure S7: Calibration Curves by Class. ECE values are 0.3643 (Normal), 0.3382 (MCI), 0.3528 (Mild), and 0.3225 (Severe).

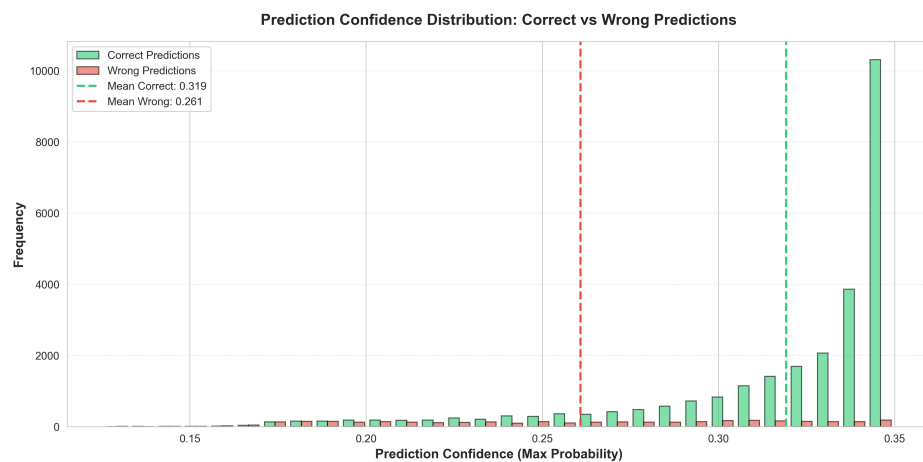


Figure S8: Confidence Distribution Analysis. Correct predictions have mean maximum probability 0.319 versus 0.261 for incorrect predictions.